

where I is the $n \times n$ identity matrix. Similarly, replacing the i th column of A by the j th column and by expanding on the i th column, we have

$$\sum_{r=1}^n a'_{ri} a_{rj} = \begin{cases} \det(A) & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (3)$$

Relation (3) yields $(A')^T A = (\det(A))I$.

The matrix $(A')^T$ is called the **adjoint** of A and is denoted by $\text{adj}(A)$. We have established an important relationship between a matrix and its adjoint.

THEOREM 4.6 Property of the Adjoint

Let A be an $n \times n$ matrix. The adjoint $\text{adj}(A) = (A')^T$ of A satisfies

$$(\text{adj}(A))A = A(\text{adj}(A)) = (\det(A))I,$$

where I is the $n \times n$ identity matrix.

Theorem 4.6 provides a formula for the inverse of an invertible matrix, which we present as a corollary.

COROLLARY A Formula for the Inverse of an Invertible Matrix

Let $A = [a_{ij}]$ be an $n \times n$ matrix with $\det(A) \neq 0$. Then A is invertible, and

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A),$$

where $\text{adj}(A) = [a'_{ij}]^T$ is the transposed matrix of cofactors.

EXAMPLE 4 Find the inverse of

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 2 & 0 \\ 3 & 1 & 1 \end{bmatrix}$$

if the matrix is invertible, using the corollary of Theorem 4.6.

SOLUTION We find that $\det(A) = 4$, so A is invertible. The cofactors a'_{ij} are

$$a'_{11} = (-1)^2 \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \quad a'_{12} = (-1)^3 \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} = -2,$$

$$a'_{13} = (-1)^4 \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} = -4, \quad a'_{21} = (-1)^3 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1,$$

$$a'_{22} = (-1)^4 \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} = 1, \quad a'_{23} = (-1)^5 \begin{vmatrix} 4 & 0 \\ 3 & 1 \end{vmatrix} = -4,$$

$$a'_{31} = (-1)^4 \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -2, \quad a'_{32} = (-1)^5 \begin{vmatrix} 4 & 1 \\ 2 & 0 \end{vmatrix} = 2,$$

$$a'_{33} = (-1)^6 \begin{vmatrix} 4 & 0 \\ 2 & 2 \end{vmatrix} = 8.$$

Hence,

$$A' = [a'_{ij}] = \begin{bmatrix} 2 & -2 & -4 \\ 1 & 1 & -4 \\ -2 & 2 & 8 \end{bmatrix}, \quad \text{so} \quad \text{adj}(A) = \begin{bmatrix} 2 & 1 & -2 \\ -2 & 1 & 2 \\ -4 & -4 & 8 \end{bmatrix}$$

and

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{4} \begin{bmatrix} 2 & 1 & -2 \\ -2 & 1 & 2 \\ -4 & -4 & 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{4} & \frac{1}{2} \\ -1 & -1 & 2 \end{bmatrix}.$$

The method described in Section 1.5 for finding the inverse of an invertible matrix is more efficient than the method illustrated in the preceding example, especially if the matrix is large. The corollary is often used to find the inverse of a 2×2 matrix. We see that if $ad - bc \neq 0$, then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

SUMMARY

1. A computationally feasible algorithm for finding the determinant of a matrix is to reduce the matrix to echelon form, using just row-addition and row-interchange operations. If a row of zeros is formed during the process, the determinant is zero. Otherwise, the determinant of the original matrix is found by computing $(-1)^r \cdot (\text{Product of pivots})$ in the echelon form, where r is the number of row interchanges performed. This is one way to program a computer to find a determinant.
2. The determinant of a matrix can be found by row or column reduction of the matrix to a matrix having a sole nonzero entry in some column or row. One then expands by minors on that column or row, and continues this process. If a matrix having a zero row or column is encountered, the determinant is zero. Otherwise, one continues until the computation is reduced to the determinant of a 2×2 matrix. This is a good way to find a determinant when working with pencil and paper.
3. If A is invertible, the linear system $Ax = b$ has the unique solution x whose k th component is given explicitly by the formula

$$x_k = \frac{\det(B_k)}{\det(A)},$$

where the matrix B_k is obtained from matrix A by replacing the k th column of A by b .

4. The methods of Chapter 1 are far more efficient than those described in this section for actual computation of both the inverse of A and the solution of the system $Ax = b$.
5. Let A be an $n \times n$ matrix, and let A' be its matrix of cofactors. The adjoint $\text{adj}(A)$ is the matrix $(A')^T$ and satisfies $(\text{adj}(A))A = A(\text{adj}(A)) = (\det(A))I$, where I is the $n \times n$ identity matrix.
6. The inverse of an invertible matrix A is given by the explicit formula

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

EXERCISES

In Exercises 1–10, find the determinant of the given matrix.

1. $\begin{bmatrix} 2 & 3 & -1 \\ 5 & -7 & 1 \\ -3 & 2 & -1 \end{bmatrix}$

2. $\begin{bmatrix} 4 & -3 & 2 \\ -1 & -1 & 1 \\ -5 & 5 & 7 \end{bmatrix}$

3. $\begin{bmatrix} 5 & 2 & 4 & 0 \\ 2 & -3 & -1 & 2 \\ 3 & -4 & 3 & 7 \\ 1 & -1 & 0 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 3 & -5 & -1 & 7 \\ 0 & 3 & 1 & -6 \\ 2 & -5 & -1 & 8 \\ -8 & 8 & 2 & -9 \end{bmatrix}$

5. $\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 3 & -1 & 2 & 0 & 0 \\ 0 & 4 & 1 & -1 & 2 \\ 0 & 0 & -3 & 2 & 4 \\ 0 & 0 & 0 & -1 & 3 \end{bmatrix}$

6. $\begin{bmatrix} 3 & 2 & 0 & 0 & 0 \\ -1 & 4 & 1 & 0 & 0 \\ 0 & -3 & 5 & 2 & 0 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}$

7. $\begin{bmatrix} 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 2 & 0 & -3 \\ 0 & -2 & 1 & 0 & 0 \\ 5 & -3 & 2 & 0 & 0 \\ -3 & 4 & 0 & 0 & 0 \end{bmatrix}$

8. $\begin{bmatrix} 2 & -1 & 0 & 0 \\ 4 & 5 & 0 & 0 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & -4 & 2 \end{bmatrix}$

9. $\begin{bmatrix} 2 & -1 & 3 & 0 & 0 \\ 0 & 1 & 4 & 0 & 0 \\ -5 & 2 & 6 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -2 & 8 \end{bmatrix}$

10. $\begin{bmatrix} 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 2 & 1 \\ -1 & 2 & 4 & 0 & 0 \\ 3 & 1 & -2 & 0 & 0 \\ 5 & 1 & 5 & 0 & 0 \end{bmatrix}$

11. The matrices in Exercises 8 and 9 have zero entries except for entries in an $r \times r$ submatrix R and a separate $s \times s$ submatrix S whose main diagonals lie on the main diagonal of the whole $n \times n$ matrix, and where $r + s = n$. Prove that, if A is such a matrix with submatrices R and S , then $\det(A) = \det(R) \cdot \det(S)$.
12. The matrix A in Exercise 10 has a structure similar to that discussed in Exercise 11, except that the square submatrices R and S lie along the other diagonal. State and prove a result similar to that in Exercise 11 for such a matrix.
13. State and prove a generalization of the result in Exercise 11, when the matrix A has zero entries except for entries in k submatrices positioned along the diagonal.

In Exercises 14–19, use the corollary to Theorem 4.6 to find A^{-1} if A is invertible.

14. $A = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$

15. $A = \begin{bmatrix} 4 & 1 \\ 2 & 1 \end{bmatrix}$

$$16. A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ -2 & 1 & 1 \end{bmatrix} \quad 17. A = \begin{bmatrix} 3 & 0 & 4 \\ -2 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$18. A = \begin{bmatrix} 3 & 0 & 3 \\ 4 & 1 & -2 \\ -5 & 1 & 4 \end{bmatrix} \quad 19. A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 1 \end{bmatrix}$$

$$20. \text{ Find the adjoint of the matrix } \begin{bmatrix} 4 & 5 \\ -3 & 6 \end{bmatrix}.$$

$$21. \text{ Find the adjoint of the matrix } \begin{bmatrix} 2 & 1 & 0 \\ 3 & 1 & 4 \\ 0 & 2 & 1 \end{bmatrix}.$$

$$22. \text{ Given that } A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } \det(A^{-1}) = 3, \text{ find the matrix } A.$$

$$23. \text{ If } A \text{ is a matrix with integer entries and if } \det(A) = \pm 1, \text{ prove that } A^{-1} \text{ also has the same properties.}$$

In Exercises 24–31, solve the given system of linear equations by Cramer's rule wherever it is possible.

$$24. \begin{cases} x_1 - 2x_2 = 1 \\ 3x_1 + 4x_2 = 3 \end{cases} \quad 25. \begin{cases} 2x_1 - 3x_2 = 1 \\ -4x_1 + 6x_2 = -2 \end{cases}$$

$$26. \begin{cases} 3x_1 + x_2 = 5 \\ 2x_1 + x_2 = 0 \end{cases} \quad 27. \begin{cases} x_1 + x_2 = 1 \\ x_1 + 2x_2 = 2 \end{cases}$$

$$28. \begin{cases} 5x_1 - 2x_2 + x_3 = 1 \\ x_2 + x_3 = 0 \\ x_1 + 6x_2 - x_3 = 4 \end{cases}$$

$$29. \begin{cases} x_1 + 2x_2 - x_3 = -2 \\ 2x_1 + x_2 + x_3 = 0 \\ 3x_1 - x_2 + 5x_3 = 1 \end{cases}$$

$$30. \begin{cases} x_1 - x_2 + x_3 = 0 \\ x_1 + 2x_2 - x_3 = 1 \\ x_1 - x_2 + 2x_3 = 0 \end{cases}$$

$$31. \begin{cases} 3x_1 + 2x_2 - x_3 = 1 \\ x_1 - 4x_2 + x_3 = -2 \\ 5x_1 + 2x_2 = 1 \end{cases}$$

In Exercises 32 and 33, find the component x_2 of the solution vector for the given linear system.

$$32. \begin{cases} x_1 + x_2 - 3x_3 + x_4 = 1 \\ 2x_1 + x_2 + 2x_4 = 0 \\ x_2 - 6x_3 - x_4 = 5 \\ 3x_1 + x_2 + x_4 = 1 \end{cases}$$

$$33. \begin{cases} 6x_1 + x_2 - x_3 = 4 \\ x_1 - x_2 + 5x_4 = -2 \\ -x_1 + 3x_2 + x_3 = 2 \\ x_1 + x_2 - x_3 + 2x_4 = 0 \end{cases}$$

$$34. \text{ Find the unique solution (assuming that it exists) of the system of equations represented by the partitioned matrix}$$

$$\left[\begin{array}{cccc|c} a_1 & b_1 & c_1 & d_1 & 3b_1 \\ a_2 & b_2 & c_2 & d_2 & 3b_2 \\ a_3 & b_3 & c_3 & d_3 & 3b_3 \\ a_4 & b_4 & c_4 & d_4 & 3b_4 \end{array} \right]$$

$$35. \text{ Let } A \text{ be a square matrix. Mark each of the following True or False.}$$

- a. The determinant of a square matrix is the product of the entries on its main diagonal.
- b. The determinant of an upper-triangular square matrix is the product of the entries on its main diagonal.
- c. The determinant of a lower-triangular square matrix is the product of the entries on its main diagonal.
- d. A square matrix is nonsingular if and only if its determinant is positive.
- e. The column vectors of an $n \times n$ matrix are independent if and only if the determinant of the matrix is nonzero.
- f. A homogeneous square linear system has a nontrivial solution if and only if the determinant of its coefficient matrix is zero.
- g. The product of a square matrix and its adjoint is the identity matrix.
- h. The product of a square matrix and its adjoint is equal to some scalar times the identity matrix.
- i. The transpose of the adjoint of A is the matrix of cofactors of A .
- j. The formula $A^{-1} = (1/\det(A))\text{adj}(A)$ is of practical use in computing the inverse of a large nonsingular matrix.

36. Prove that the inverse of a nonsingular upper-triangular matrix is upper triangular.
37. Prove that a square matrix is invertible if and only if its adjoint is an invertible matrix.
38. Let A be an $n \times n$ matrix. Prove that $\det(\text{adj}(A)) = \det(A)^{n-1}$.
39. Let A be an invertible $n \times n$ matrix with $n > 1$. Using Exercises 37 and 38, prove that $\text{adj}(\text{adj}(A)) = (\det(A))^{n-2}A$.

 The routine YUREDUCE in LINTEK has a menu option D that will compute and display the product of the diagonal elements of a square matrix. The routine MATCOMP has a menu option D to compute a determinant. Use YUREDUCE or MATLAB to compute the determinant of the matrices in Exercises 40–42. Write down your results. If you used YUREDUCE, use MATCOMP to compute the determinants of the same matrices again and compare the answers.

40. $\begin{bmatrix} 11 & -9 & 28 \\ 32 & -24 & 21 \\ 10 & 13 & -19 \end{bmatrix}$ 41. $\begin{bmatrix} 13 & -15 & 33 \\ -15 & 25 & 40 \\ 12 & -33 & 27 \end{bmatrix}$

42. $\begin{bmatrix} 7.6 & 2.8 & -3.9 & 19.3 & 25.0 \\ -33.2 & 11.4 & 13.2 & 22.4 & 18.3 \\ 21.4 & -32.1 & 45.7 & -8.9 & 12.5 \\ 17.4 & 11.0 & -6.8 & 20.3 & -35.1 \\ 22.7 & 11.9 & 33.2 & 2.5 & 7.8 \end{bmatrix}$

43. MATCOMP computes determinants in essentially the way described in this section. The matrix

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$$

has determinant 1, so every power of it should have determinant 1. Use MATCOMP with single-precision printing and with the

default roundoff control ratio r . Start computing determinants of powers of A . Find the smallest positive integer m such that $\det(A^m) \neq 1$, according to MATCOMP. How bad is the error? What does MATCOMP give for $\det(A^{20})$? At what integer exponent does the break occur between the incorrect value 0 and incorrect values of large magnitude?

Repeat the above, taking zero for roundoff control ratio r . Try to explain why the results are different and what is happening in each case.

44. Using MATLAB, find the smallest positive integer m such that $\det(A^m) \neq 1$, according to MATLAB, for the matrix A in Exercise 43.

In Exercises 45–47, use MATCOMP in LINTEK or MATLAB and the corollary of Theorem 4.6 to find the matrix of cofactors of the given matrix.

45. $\begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 0 \\ 3 & 1 & 4 \end{bmatrix}$

46. $\begin{bmatrix} -52 & 31 & 47 \\ 21 & -11 & 28 \\ 43 & -71 & 87 \end{bmatrix}$

47. $\begin{bmatrix} 6 & -3 & 2 & 14 \\ -3 & 7 & 8 & 1 \\ 4 & 9 & -5 & 3 \\ -8 & -40 & 47 & 29 \end{bmatrix}$

[Hint: Entries in the matrix of cofactors are integers. The cofactors of a matrix are continuous functions of its entries; that is, changing an entry by a very slight amount will change a cofactor only slightly. Change some entry just a bit to make the determinant nonzero.]

4.4

LINEAR TRANSFORMATIONS AND DETERMINANTS (OPTIONAL)

We continue our program of exhibiting the relationship between matrices and linear transformations. Associated with an $m \times n$ matrix A is the linear transformation T mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, where $T(x) = Ax$ for x in $\mathbb{R}^n$. If $n = m$, so

that A is a square matrix, then $\det(A)$ is defined. But what is the meaning of the number $\det(A)$ for the transformation T ? We now tackle this question, and the answer is so important that it merits a section all to itself. The notion of the determinant associated with a linear transformation T mapping $\mathbb{R}^n$ into $\mathbb{R}^n$ lies at the heart of variable substitution in integral calculus. This section presents an informal and intuitive explanation of this notion.

The Volume of an n -Box in $\mathbb{R}^m$

In Section 4.1, we saw that the area of the parallelogram (or *2-box*) in $\mathbb{R}^2$ determined by vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ is the absolute value $|\det(A)|$ of the determinant of the 2×2 matrix A having $\mathbf{a}_1$ and $\mathbf{a}_2$ as column vectors.* We also saw that the volume of the parallelepiped (or *3-box*) in $\mathbb{R}^3$ determined by vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ is $|\det(A)|$ for the 3×3 matrix A whose column vectors are $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$. We wish to extend these notions by defining an n -box in $\mathbb{R}^m$ for $m \geq n$ and finding its "volume."

DEFINITION 4.2 An n -Box in $\mathbb{R}^m$

Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be n independent vectors in $\mathbb{R}^m$ for $n \leq m$. The n -box in $\mathbb{R}^m$ determined by these vectors is the set of all vectors $\mathbf{x}$ satisfying

$$\mathbf{x} = t_1\mathbf{a}_1 + t_2\mathbf{a}_2 + \cdots + t_n\mathbf{a}_n$$

for $0 \leq t_i \leq 1$ and $i = 1, 2, \dots, n$.

If the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ in Definition 4.2 are dependent, the set described is a *degenerate* n -box.

EXAMPLE 1 Describe geometrically the 1-box determined by the "vector" 2 in $\mathbb{R}$ and the 1-box determined by a nonzero vector $\mathbf{a}$ in $\mathbb{R}^m$.

SOLUTION The 1-box determined by the "vector" 2 in $\mathbb{R}$ consists of all numbers $t(2)$ for $0 \leq t \leq 1$, which is simply the closed interval $0 \leq x \leq 2$. Similarly, the 1-box in $\mathbb{R}^m$ determined by a nonzero vector $\mathbf{a}$ is the line segment joining the origin to the tip of $\mathbf{a}$. ■

EXAMPLE 2 Draw symbolic sketches of a 2-box in $\mathbb{R}^m$ and a 3-box in $\mathbb{R}^m$.

SOLUTION A 2-box in $\mathbb{R}^m$ is a parallelogram with a vertex at the origin, as shown in Figure 4.7. Similarly, a 3-box in $\mathbb{R}^m$ is a parallelepiped with a vertex at the origin, as illustrated in Figure 4.8. ■

Notice that our boxes need not be rectangular boxes; perhaps we should have used the term *skew box* to make this clear.

*Anticipating our work in this section, we arrange the vectors $\mathbf{a}_i$ as columns rather than as rows of A . Recall that $\det(A) = \det(A^T)$.

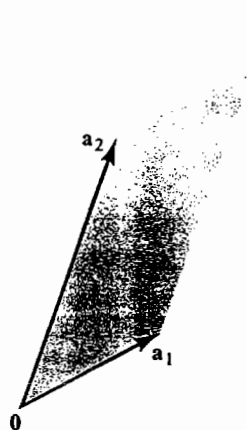


FIGURE 4.7
A 2-box in $\mathbb{R}^m$.

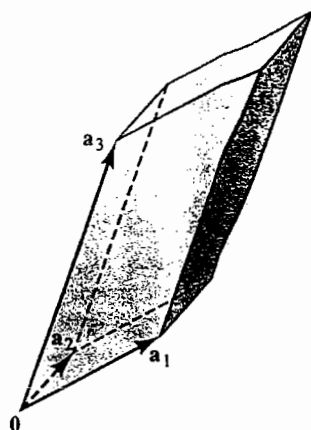


FIGURE 4.8
A 3-box in $\mathbb{R}^m$.

We are accustomed to speaking of the *length* of a line segment, of the *area* of a piece of the plane, and of the *volume* of a piece of space. To avoid introducing new terms when discussing a general n -box, we will use the three-dimensional term *volume* when speaking of its size. Notice that we already used the three-dimensional term *box* as the name for the object! Thus, by the *volume* of a 1-box, we simply mean its length; the *volume* of a 2-box is its area, and so on.

The volume of the 1-box determined by $\mathbf{a}_1$ in $\mathbb{R}^m$ is its length $\|\mathbf{a}_1\|$. Because the determinant of a 1×1 matrix is its sole entry, Exercise 36 shows that this length can be written as

$$\text{Length} = \|\mathbf{a}_1\| = \sqrt{\det([\mathbf{a}_1 \cdot \mathbf{a}_1])}. \quad (1)$$

Let us turn to a 2-box in $\mathbb{R}^m$ determined by nonzero and nonparallel vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. We repeat an argument that we made in Section 4.1 for a 2-box in $\mathbb{R}^2$, using a slightly different notation for this general m . From Figure 4.9, we see that the area of this parallelogram is given by

$$\text{Area} = \|h\| \|\mathbf{a}_2\|,$$

where, for the angle θ between $\mathbf{a}_1$ and $\mathbf{a}_2$, we have $\|h\| = \|\mathbf{a}_1\| \sin \theta$. We then have

$$\begin{aligned} (\text{Area})^2 &= \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 \sin^2 \theta \\ &= \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 (1 - \cos^2 \theta) \\ &= \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - (\|\mathbf{a}_1\| \|\mathbf{a}_2\| \cos \theta)^2 \\ &= (\mathbf{a}_1 \cdot \mathbf{a}_1)(\mathbf{a}_2 \cdot \mathbf{a}_2) - (\mathbf{a}_1 \cdot \mathbf{a}_2)(\mathbf{a}_2 \cdot \mathbf{a}_1) \\ &= \begin{vmatrix} \mathbf{a}_1 \cdot \mathbf{a}_1 & \mathbf{a}_1 \cdot \mathbf{a}_2 \\ \mathbf{a}_2 \cdot \mathbf{a}_1 & \mathbf{a}_2 \cdot \mathbf{a}_2 \end{vmatrix} = \det([\mathbf{a}_1 \cdot \mathbf{a}_2]). \end{aligned} \quad (2)$$

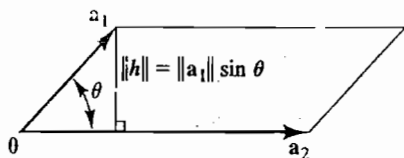


FIGURE 4.9
The volume of a 2-box in $\mathbb{R}^m$ is
(Length of the base) $\times$ (Altitude).

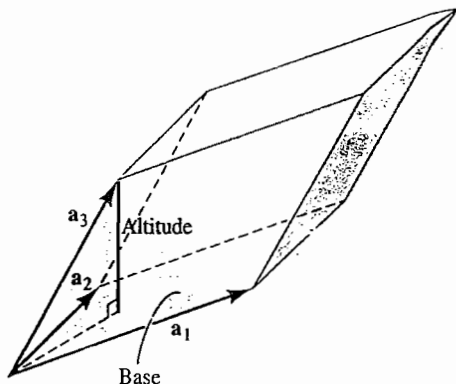


FIGURE 4.10
The volume of a 3-box in $\mathbb{R}^m$ is
(Area of the base) $\times$ (Altitude).

From Eqs. (1) and (2), we might guess that the square of the volume of an n -box in $\mathbb{R}^m$ is $\det([a_i \cdot a_j])$. Of course, we must define what we mean by the volume of such a box, but with the natural definition, this conjecture is true. If A is the matrix with j th column vector a_j , then A^T is the matrix with i th row vector a_i , and the $n \times n$ matrix $[a_i \cdot a_j]$ is $A^T A$, so Eq. (2) can be written as

$$(\text{Area}) = \sqrt{\det(A^T A)}.$$

We have an intuitive idea of the volume of an n -box in $\mathbb{R}^m$ for $n \leq m$. For example, the 3-box in $\mathbb{R}^m$ determined by independent vectors a_1, a_2, a_3 has a volume equal to the altitude of the box times the volume (that is, area) of the base, as shown in Figure 4.10. Roughly speaking, the volume of an n -box is equal to the altitude of the box times the volume of the base, which is an $(n-1)$ -box. This notion of the *altitude* of a box can be made precise after we develop projections in Chapter 6. The formal definition of the volume of an n -box appears in Appendix B, as does the proof of our main result on volumes (Theorem 4.7). For the remainder of this section, we are content to proceed with our intuitive notion of volume.

THEOREM 4.7 Volume of a Box

The volume of the n -box in $\mathbb{R}^m$ determined by independent vectors $a_1, a_2, \dots, a_n$ is given by

$$\text{Volume} = \sqrt{\det(A^T A)},$$

where A is the $m \times n$ matrix with a_j as j th column vector.

The volume of an n -box in $\mathbb{R}^n$ is of such importance that we restate this special case as a corollary.

COROLLARY Volume of an n -Box in $\mathbb{R}^n$

If A is an $n \times n$ matrix with independent column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, then $|\det(A)|$ is the volume of the n -box in $\mathbb{R}^n$ determined by these n vectors.

PROOF By Theorem 4.7, the square of the volume of the n -box is $\det(A^T A)$. But because A is an $n \times n$ matrix, we have

$$\det(A^T A) = \det(A^T) \cdot \det(A) = (\det(A))^2.$$

The conclusion of the corollary then follows at once. $\blacktriangle$

EXAMPLE 3 Find the area of the parallelogram in $\mathbb{R}^4$ determined by the vectors $[2, 1, -1, 3]$ and $[0, 2, 4, -1]$.

SOLUTION If

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 2 \\ -1 & 4 \\ 3 & -1 \end{bmatrix},$$

then

$$A^T A = \begin{bmatrix} 2 & 1 & -1 & 3 \\ 0 & 2 & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 2 \\ -1 & 4 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 15 & -5 \\ -5 & 21 \end{bmatrix}.$$

By Theorem 4.7, we have

$$(\text{Area})^2 = \begin{vmatrix} 15 & -5 \\ -5 & 21 \end{vmatrix} = 290.$$

Thus the area of the parallelogram is $\sqrt{290}$. $\blacksquare$

EXAMPLE 4 Find the volume of the parallelepiped in $\mathbb{R}^3$ determined by the vectors $[1, 0, -1]$, $[-1, 1, 3]$, and $[2, 4, 1]$.

SOLUTION We compute the determinant

$$\begin{vmatrix} 1 & -1 & 2 \\ 0 & 1 & 4 \\ -1 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 2 \\ 0 & 1 & 4 \\ 0 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = -5.$$

Applying the corollary of Theorem 4.7, the volume of the parallelepiped is therefore 5. $\blacksquare$

Comparing Theorem 4.7 and its corollary, we see that the formula for the volume of an n -box in a space of larger dimension m involves a square root,

whereas the formula for the volume of a box in a space of its own dimension does not involve a square root. The student of calculus discovers that the calculus formulas used to find the length of a curve (which is one-dimensional) in the plane or in space involve a square root. The same is true of the formulas used to find the area of a surface (two-dimensional) in space. However, the calculus formulas for finding the area of part of the plane or the volume of some part of space do not involve square roots. Theorem 4.7 and its corollary lie at the heart of this difference in the calculus formulas.

Volume-Change Factor of a Linear Transformation

Recall the multiplicative property of determinants: $\det(AB) = \det(A) \cdot \det(B)$, where A and B are $n \times n$ matrices. Suppose that both A and B have rank n . Then the column vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ of B determine an n -box having volume $|\det(B)| \neq 0$. From the definition of matrix multiplication, the j th column vector of AB is $A\mathbf{b}_j$, and we write

$$AB = \begin{bmatrix} | & | & & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \\ | & | & & | \end{bmatrix}.$$

The linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$ thus carries the original n -box determined by the column vectors of B into a new n -box determined by the column vectors of AB . The new n -box has volume $|\det(AB)| = |\det(A)| \cdot |\det(B)|$. That is, the volume of the new n -box, or *image box*, is equal to $|\det(A)|$ times the volume of the original box. Thus $|\det(A)|$ is referred to as the **volume-change factor** for the linear transformation T . We are interested in this concept only when $\det(A) \neq 0$, a requirement that ensures that A is invertible, that the n vectors $A\mathbf{b}_j$ are independent, and that T is an invertible transformation.

To illustrate this idea of the volume-change factor, consider the n -cube in $\mathbb{R}^n$ determined by the vectors $c\mathbf{e}_1, c\mathbf{e}_2, \dots, c\mathbf{e}_n$ for $c > 0$. This n -cube with edges of length c has volume c^n . It is carried by T into an n -box having volume $|\det(A)| \cdot c^n$; the image n -box need not be a cube, nor even rectangular. We illustrate the image n -box in Figure 4.11 for the case $n = 1$ and in Figure 4.12 for the case $n = 3$.

EXAMPLE 5 Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T([x_1, x_2, x_3]) = [x_1 + x_3, 2x_2, 2x_1 + 5x_3].$$

Find the volume of the image box when T acts on the cube determined by the vectors $c\mathbf{e}_1, c\mathbf{e}_2$, and $c\mathbf{e}_3$ for $c > 0$.

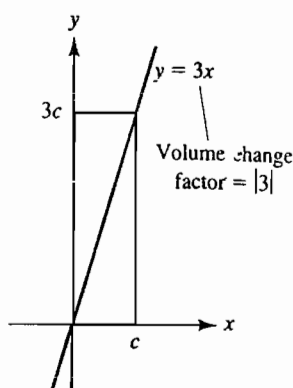


FIGURE 4.11

The volume-change factor of $T(x) = 3x$.

SOLUTION The image box is determined by the vectors

$$T(ce_1) = cT(e_1) = c[1, 0, 2]$$

$$T(ce_2) = cT(e_2) = c[0, 2, 0]$$

$$T(ce_3) = cT(e_3) = c[1, 0, 5].$$

The standard matrix representation of T is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 2 & 0 & 5 \end{bmatrix},$$

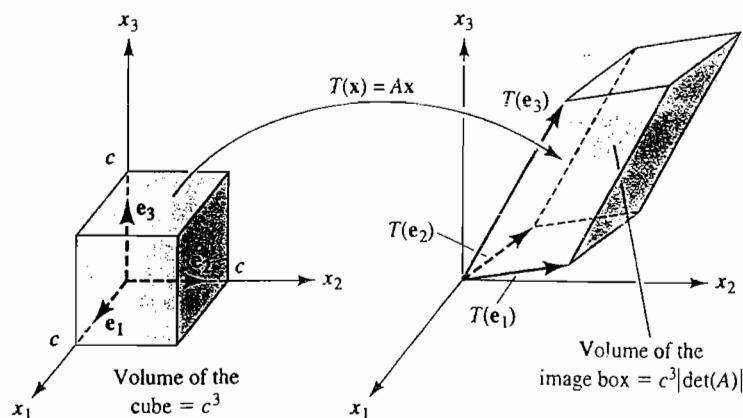


FIGURE 4.12

The volume-change factor of $T(x) = Ax$.

and the volume-change factor of T is $|\det(A)| = 6$. Therefore, the image box has volume $6c^3$. This volume can also be computed by taking the determinant of the matrix having as column vectors $T(ce_1)$, $T(ce_2)$, and $T(ce_3)$. This matrix is cA , and has determinant $c^3 \cdot \det(A) = 6c^3$. ■

The sign of the determinant associated with an invertible linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ depends on whether T preserves orientation. In the plane where $n = 2$, orientation is preserved by T if $T(e_1)$ can be rotated counterclockwise through an angle of less than 180° to lie along $T(e_2)$. It can be shown that this is the case if and only if $\det(A) > 0$, where A is the standard matrix representation of T . In general, a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to *preserve orientation* if the determinant of its standard matrix representation A is positive. Thus the linear transformation in Example 5 preserves orientation because $\det(A) = 6$. Because this topic is more suitable for a course in analysis than for one in linear algebra, we do not pursue it further.

Application to Calculus

We can get an intuitive feel for the connection between the volume-change factor of T and integral calculus. The definition of an integral involves summing products of the form

$$\underbrace{(\text{Function value at some point of a box})}_{f(x)} \underbrace{(\text{Volume of the box})}_{dx}. \quad (3)$$

Under a change of variables—say, from x -variables to t -variables—the boxes in the dx -space are replaced by boxes in dt -space via an invertible linear transformation—namely, the *differential* of the variable substitution function. Thus the second factor in product (3) must be expressed in terms of volumes of boxes in the dt -space. The determinant of the differential transformation must play a role, because the volume of the box in dx -space is the volume of the corresponding box in dt -space multiplied by the absolute value of the determinant.

Let us look at a one-dimensional example. In making the substitution $x = \sin t$ in an integral in single-variable calculus, we associate with each t -value t_0 the linear transformation of dt -space into dx -space given by the equation $dx = (\cos t_0)dt$. The determinant of this linear transformation is $\cos t_0$. A little 1-box of volume (length) dt and containing the point t_0 is carried by this linear transformation into a little 1-box in the dx -space of volume (length) $|\cos t_0|dt$. Having conveyed a rough idea of this topic and its importance, we leave its further development to an upper-level course in analysis.

Regions More General Than Boxes

In the remainder of this section, we will be working with the volume of *sufficiently nice regions* in $\mathbb{R}^n$. To define such a notion carefully would require an excursion into calculus of several variables. We will simply assume that we all have an intuitive notion of such regions having a well-defined volume.

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation of rank n , and let A be its standard matrix representation. Recall that the j th column vector of A is $T(\mathbf{e}_j)$ (see p. 146). We will show that, if a region G in $\mathbb{R}^n$ has volume V , the image of G under T has volume

$$|\det(A)| \cdot V.$$

That is, the volume of a region is multiplied by $|\det(A)|$ when the region is transformed by T . This result has important applications to integral calculus.

The volume of a sufficiently nice region G in $\mathbb{R}^n$ may be approximated by adding the volumes of small n -cubes contained in G and having edges of length c parallel to the coordinate axes. Figure 4.13(a) illustrates this situation for a plane region in $\mathbb{R}^2$, where a grid of small squares (2-cubes) is placed on the region. As the length c of the edges of the squares approaches zero, the sum of the areas of the colored squares inside the region approaches the area of the region. These squares inside G are mapped by T into parallelograms of area $c^2|\det(A)|$ inside the image of G under T . (See the colored parallelograms in Figure 4.13(b). As c approaches zero, the sum of the areas of these parallelograms approaches the area of the image of G under T , which thus must be the area of G multiplied by $|\det(A)|$. A similar construction can be made with a grid of n -cubes for a region G in $\mathbb{R}^n$. Each such cube is mapped by T into an n -box of volume $c^n|\det(A)|$. Adding the volumes of these n -boxes and taking the limiting value of the sum as c approaches zero, we see that the volume of the image under T of the region G is given by

$$\text{Volume of image of } G = |\det(A)| \cdot (\text{Volume of } G). \quad (4)$$

We summarize this work in a theorem on the following page.

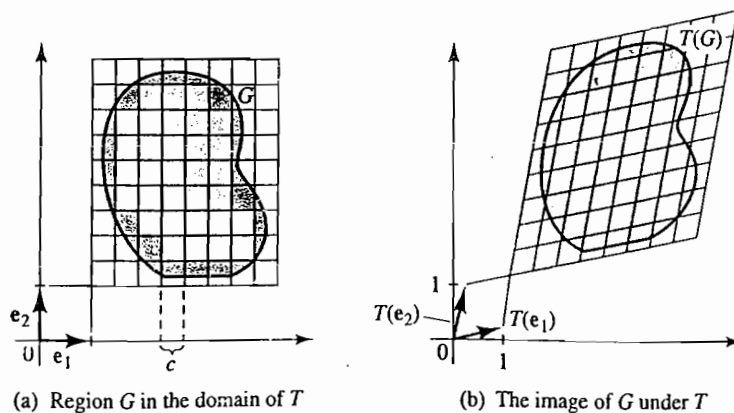


FIGURE 4.13

THEOREM 4.8 Volume-Change Factor for $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$

Let G be a region in $\mathbb{R}^n$ of volume V , and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation of rank n with standard matrix representation A . Then the volume of the image of G under T is $|\det(A)| \cdot V$.

EXAMPLE 6 Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation of the plane given by $T([x, y]) = [2x - y, x + 3y]$. Find the area of the image under T of the disk $x^2 + y^2 \leq 4$ in the domain of T .

SOLUTION The disk $x^2 + y^2 \leq 4$ has radius 2 and area 4π . It is mapped by T into a region (actually bounded by an ellipse) of area

$$|\det(A)| \cdot (4\pi) = \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} \cdot (4\pi) = (6 + 1)(4\pi) = 28\pi. \quad \blacksquare$$

We can generalize Theorem 4.8 to a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, where $m \geq n$ and T has rank n . This time, the standard matrix representation A is an $m \times n$ matrix. The image under T of the unit n -cube in $\mathbb{R}^n$ outlined by $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is the n -box in $\mathbb{R}^m$ outlined by $T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$. According to Theorem 4.7, the volume of this box in $\mathbb{R}^m$ is

$$\sqrt{\det(A^T A)}.$$

The same grid argument used earlier and illustrated in Figure 4.13 shows that a region G in $\mathbb{R}^n$ of volume V is mapped by T into a region of $\mathbb{R}^m$ of volume

$$\sqrt{\det(A^T A)} \cdot V.$$

We summarize this generalization in a theorem.

THEOREM 4.9 Volume-Change Factor for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

Let G be a region in $\mathbb{R}^n$ of volume V . Let $m \geq n$ and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation of rank n . Let A be the standard matrix representation of T . Then the volume of the image of G in $\mathbb{R}^m$ under the transformation T is

$$\sqrt{\det(A^T A)} \cdot V.$$

EXAMPLE 7 Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be given by $T([x, y]) = [2x + 3y, x - y, 2y]$. Find the area of the image in $\mathbb{R}^3$ under T of the disk $x^2 + y^2 \leq 4$ in $\mathbb{R}^2$.

SOLUTION The standard matrix representation A of T is

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \\ 0 & 2 \end{bmatrix}$$

and

$$A^T A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 14 \end{bmatrix}.$$

Thus,

$$\sqrt{\det(A^T A)} = \sqrt{70 - 25} = \sqrt{45} = 3\sqrt{5}.$$

A region G of $\mathbb{R}^2$ having area V is mapped by T into a plane region of area $3\sqrt{5} \cdot V$ in $\mathbb{R}^3$. Thus the disk $x^2 + y^2 \leq 4$ of area 4π is mapped into a plane region in $\mathbb{R}^3$ of area

$$(3\sqrt{5})(4\pi) = 12\pi\sqrt{5}.$$

SUMMARY

1. An n -box in $\mathbb{R}^m$, where $m \geq n$, is determined by n independent vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ and consists of all vectors $\mathbf{x}$ in $\mathbb{R}^m$ such that

$$\mathbf{x} = t_1 \mathbf{a}_1 + t_2 \mathbf{a}_2 + \dots + t_n \mathbf{a}_n,$$

where $0 \leq t_i \leq 1$ for $i = 1, 2, \dots, n$.

2. A 1-box in $\mathbb{R}^m$ is a line segment, and its "volume" is its length.
3. A 2-box in $\mathbb{R}^m$ is a parallelogram determined by two independent vectors, and the "volume" of the 2-box is the area of the parallelogram.
4. A 3-box in $\mathbb{R}^m$ is a skewed box (parallelepiped) in the usual sense, and its volume is the usual volume.
5. Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be independent vectors in $\mathbb{R}^m$ for $m \geq n$, and let A be the $m \times n$ matrix with j th column vector $\mathbf{a}_j$. The volume of the n -box in $\mathbb{R}^m$ determined by the n vectors is $\sqrt{\det(A^T A)}$.
6. For the case of an n -box in the space $\mathbb{R}^n$ of the same dimension, the formula for its volume given in summary item 5 reduces to $|\det(A)|$.
7. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation of rank n with standard matrix representation A , then T maps a region in its domain of volume V into a region of volume $|\det(A)|V$.
8. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation of rank n with standard matrix representation A , then T maps a region in its domain of volume V into a region of $\mathbb{R}^m$ of volume $\sqrt{\det(A^T A)} \cdot V$.

EXERCISES

1. Find the area of the parallelogram in $\mathbb{R}^3$ determined by the vectors $[0, 1, 4]$ and $[-1, 3, -2]$.
2. Find the area of the parallelogram in $\mathbb{R}^5$ determined by the vectors $[1, 0, 1, 2, -1]$ and $[0, 1, -1, 1, 3]$.

3. Find the volume of the 3-box in $\mathbb{R}^4$ determined by the vectors $[-1, 2, 0, 1]$, $[0, 1, 3, 0]$, and $[0, 0, 2, -1]$.
4. Find the volume of the 4-box in $\mathbb{R}^5$ determined by the vectors $[1, 1, 1, 0, 1]$, $[0, 1, 1, 0, 0]$, $[3, 0, 1, 0, 0]$, and $[1, -1, 0, 0, 1]$.

In Exercises 5–10, find the volume of the n -box determined by the given vectors in $\mathbb{R}^n$.

5. $[-1, 4]$, $[2, 3]$ in $\mathbb{R}^2$
6. $[-5, 3]$, $[1, 7]$ in $\mathbb{R}^2$
7. $[1, 3, -5]$, $[2, 4, -1]$, $[3, 1, 2]$ in $\mathbb{R}^3$
8. $[-1, 4, 7]$, $[3, -2, -1]$, $[4, 0, 2]$ in $\mathbb{R}^3$
9. $[1, 0, 0, 1]$, $[2, -1, 3, 0]$, $[0, 1, 3, 4]$, $[-1, 1, -2, 1]$ in $\mathbb{R}^4$
10. $[1, -1, 0, 1]$, $[2, -1, 3, 1]$, $[-1, 4, 2, -1]$, $[0, 1, 0, 2]$ in $\mathbb{R}^4$
11. Find the area of the triangle in $\mathbb{R}^3$ with vertices $(-1, 2, 3)$, $(0, 1, 4)$, and $(2, 1, 5)$. [HINT: Think of vectors emanating from $(-1, 2, 3)$. The triangle may be viewed as half a parallelogram.]
12. Find the volume of the tetrahedron in $\mathbb{R}^3$ with vertices $(1, 0, 3)$, $(-1, 2, 4)$, $(3, -1, 2)$, and $(2, 0, -1)$. [HINT: Think of vectors emanating from $(1, 0, 3)$.]
13. Find the volume of the tetrahedron in $\mathbb{R}^4$ with vertices $(1, 0, 0, 1)$, $(-1, 2, 0, 1)$, $(3, 0, 1, 1)$, and $(-1, 4, 0, 1)$. [HINT: See the hint for Exercise 12.]
14. Give a geometric interpretation of the fact that an $n \times n$ matrix with two equal rows has determinant zero.
15. Using the results of this section, give a criterion that four points in $\mathbb{R}^n$ lie in a plane.
16. Determine whether the points $(1, 0, 1, 0)$, $(-1, 1, 0, 1)$, $(0, 1, -1, 1)$, and $(1, -1, 4, -1)$ lie in a plane in $\mathbb{R}^4$. (See Exercise 15.)
17. Determine whether the points $(2, 0, 1, 3)$, $(3, 1, 0, 1)$, $(-1, 2, 0, 4)$, and $(3, 1, 2, 4)$ lie in a plane in $\mathbb{R}^4$. (See Exercise 15.)

In Exercises 18–21, let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation defined by $T([x, y]) =$

$[4x - 2y, 2x + 3y]$. Find the area of the image under T of each of the given regions in $\mathbb{R}^2$.

18. The square $0 \leq x \leq 1, 0 \leq y \leq 1$
19. The rectangle $-1 \leq x \leq 1, 1 \leq y \leq 2$
20. The parallelogram determined by $2\mathbf{e}_1 + 3\mathbf{e}_2$ and $4\mathbf{e}_1 - \mathbf{e}_2$
21. The disk $(x - 1)^2 + (y + 2)^2 \leq 9$

In Exercises 22–25, let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T([x, y, z]) = [x - 2y, 3x + z, 4x + 3y]$. Find the volume of the image under T of each of the given regions in $\mathbb{R}^3$.

22. The cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$
23. The box $0 \leq x \leq 2, -1 \leq y \leq 3, 2 \leq z \leq 5$
24. The box determined by $2\mathbf{e}_1 + 3\mathbf{e}_2 - \mathbf{e}_3$, $4\mathbf{e}_1 - 2\mathbf{e}_3$, and $\mathbf{e}_1 - \mathbf{e}_2 + 2\mathbf{e}_3$
25. The ball $x^2 + (y - 3)^2 + (z + 2)^2 \leq 16$

In Exercises 26–29, let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation defined by $T([x, y]) = [y, x, x + y]$. Find the area of the image under T of each of the given regions in $\mathbb{R}^2$.

26. The square $0 \leq x \leq 1, 0 \leq y \leq 1$
27. The rectangle $2 \leq x \leq 3, -1 \leq y \leq 4$
28. The triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 3)$
29. The disk $x^2 + y^2 \leq 25$

In Exercises 30–32, let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ be defined by $T([x, y]) = [x - y, x, -y, 2x + y]$. Find the area of the image under T of each of the given regions in $\mathbb{R}^2$.

30. The square $0 \leq x \leq 1, 0 \leq y \leq 1$
31. The square $-1 \leq x \leq 3, -1 \leq y \leq 3$
32. The disk $x^2 + y^2 \leq 9$
33. a. If one attempts to define an n -box in $\mathbb{R}^m$ for $n > m$, what will its volume as an n -box be?
b. Let A be an $m \times n$ matrix with $n > m$. Find $\det(A^T A)$.

34. We have seen that, for $n \times n$ matrices A and B , we have $\det(AB) = \det(A) \cdot \det(B)$, but the proof was not intuitive. Give an intuitive

geometric argument showing that at least $|\det(AB)| = |\det(A)| \cdot |\det(B)|$. [HINT: Use the fact that, if A is the standard matrix representation of $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and B is the standard matrix representation of $T': \mathbb{R}^n \rightarrow \mathbb{R}^n$, then AB is the standard matrix representation $T \circ T'$.]

35. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation of rank n with standard matrix representation A . Mark each of the following True or False.

- a. The image under T of a box in $\mathbb{R}^n$ is again a box in $\mathbb{R}^n$.
- b. The image under T of an n -box in $\mathbb{R}^n$ of volume V is a box in $\mathbb{R}^n$ of volume $\det(A) \cdot V$.
- c. The image under T of an n -box in $\mathbb{R}^n$ of volume > 0 is a box in $\mathbb{R}^n$ of volume > 0 .
- d. If the image under T of an n -box B in $\mathbb{R}^n$ has volume 12, the box B has volume $|\det(A)| \cdot 12$.

- e. If the image under T of an n -box B in $\mathbb{R}^n$ has volume 12, the box B has volume $12/|\det(A)|$.
 - f. If $n = 2$, the image under T of the unit disk $x^2 + y^2 \leq 1$ has area $|\det(A)|$.
 - g. The linear transformation T is an isomorphism.
 - h. The image under $T \circ T$ of an n -box in $\mathbb{R}^n$ of volume V is a box in $\mathbb{R}^n$ of volume $\det(A^2) \cdot V$.
 - i. The image under $T \circ T \circ T$ of an n -box in $\mathbb{R}^n$ of volume V is a box in $\mathbb{R}^n$ of volume $\det(A^3) \cdot V$.
 - j. The image under T of a nondegenerate 1-box is again nondegenerate.
36. Prove Eq. (1); that is, prove that the square of the length of the line segment determined by $\mathbf{a}_1$ in $\mathbb{R}^n$ is $\|\mathbf{a}_1\|^2 = \det([\mathbf{a}_1 \cdot \mathbf{a}_1])$.

EIGENVALUES AND
EIGENVECTORS

This chapter introduces the important topic of eigenvalues and eigenvectors of a square matrix and the associated linear transformation. Determinants enable us to give illustrative computations and applications involving matrices of very small size. Eigenvectors and eigenvalues continue to appear at intervals throughout much of the rest of the text. In Section 8.4 we discuss some other methods for computing them.

5.1

EIGENVALUES AND EIGENVECTORS

Encounters with $A^k x$

In Section 1.7 we studied Markov chains dealing with the distribution of a population among states, measured over evenly spaced time intervals. An $n \times n$ transition matrix T describes the movement of the population among the states during one time interval. The matrix T has the property that its entries are nonnegative and the sum of the entries in any column is 1. Suppose that $\mathbf{p}$ is the initial population distribution vector—that is, the column vector whose i th component is the proportion of the population in the i th state at the start of the process. Then $T\mathbf{p}$ is the corresponding population distribution vector after one time interval. Similarly, $T^2\mathbf{p}$ is the population distribution vector after two time intervals, and in general, $T^k\mathbf{p}$ gives the distribution of the population among the states after k time intervals.

Markov chains provide one example in which we are interested in computing $A^k x$ for an $n \times n$ matrix A and a column vector x of n components. We give a famous classical problem that provides another illustration.

EXAMPLE 1 (*Fibonacci's rabbits*) Suppose that newly born pairs of rabbits produce offspring during the first month of their lives, but each pair produces one pair each subsequent month. Starting with $F_1 = 1$ newly born pair in the

month, find the number F_k of pairs in the k th month, assuming that no rabbit dies.

SOLUTION In the k th month, the number of pairs of rabbits is

$$F_k = (\text{Number of pairs alive the preceding month}) \\ + (\text{Number of newly born pairs for the } k\text{th month}).$$

Because our rabbits do not produce offspring during the first month of their lives, we see that the number of newly born pairs for the k th month is the number F_{k-2} of pairs alive two months before. Thus we can write the equation above as

$$F_k = F_{k-1} + F_{k-2} \quad \text{Fibonacci's relation} \quad (1)$$

It is convenient to set $F_0 = 0$, denoting 0 pairs for month 0 before the arrival of the first newly born pair, which is presumably a gift. Thus the sequence

$$F_0, F_1, F_2, \dots, F_k, \dots$$

for the number of pairs of rabbits becomes the **Fibonacci sequence**

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots, \quad (2)$$

where each term starting with $F_2 = 0 + 1 = 1$ is the sum of the two preceding terms. For any particular k , we could compute F_k by writing out the sequence far enough. ■

Fibonacci published this problem early in the thirteenth century. The Fibonacci sequence (2) occurs naturally in a surprising number of places. For example, leaves appear in a spiral pattern along a branch. Some trees have five growths of leaves for every two turns, others have eight growths for every three turns, and still others have 13 growths for every five turns; note the appearance of these numbers in the sequence (2). A mathematical journal, the *Fibonacci Quarterly*, has published many papers relating to the Fibonacci sequence.

We said in Example 1 that F_k can be found by simply writing out enough terms of the sequence (2). That can be a tedious task, even if we want to compute only F_{30} . Linear algebra gives us another approach to this problem. The Fibonacci relation (1) can be expressed in matrix form. We see that

$$\begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k-1} \\ F_{k-2} \end{bmatrix}.$$

Thus, if we set

$$\mathbf{x}_k = \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix},$$

we find that

$$\mathbf{x}_k = A\mathbf{x}_{k-1}. \quad (3)$$

Applying Eq. (3) repeatedly, we see that

$$x_2 = Ax_1, \quad x_3 = Ax_2 = A^2x_1, \quad x_4 = Ax_3 = A^3x_1,$$

and in general

$$x_k = A^{k-1}x_1 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{k-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (4)$$

Thus we can compute the k th Fibonacci number F_k by finding A^{k-1} and multiplying it on the right by the column vector x_1 . Raising a matrix to a power is also a bit of a job, but the routine MATCOMP (in LINTEK) or MATLAB can easily find F_{30} for us. (See Exercise 45.)

Both Markov chains and the Fibonacci sequence lead us to computations of the form A^kx . Other examples leading to A^kx abound in the physical and social sciences.

Computations of A^kx arise in any process in which information given by a column vector gives rise to analogous information at a later time by multiplying the vector by a matrix A .

Eigenvalues and Eigenvectors

Suppose that A is an $n \times n$ matrix and v is a column vector with n components such that

$$Av = \lambda v \quad (5)$$

for some scalar λ . Geometrically, Eq. (5) asserts that Av is a vector parallel to v . From $Av = \lambda v$, we obtain

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2v.$$

It is easy to show, in general, that $A^k v = \lambda^k v$. (See Exercise 27.) Thus, $A^k x$ is easily computed if x is equal to *this vector* v .

DEFINITION 5.1 Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix. A scalar λ is an **eigenvalue** of A if there is a **nonzero** column vector v in $\mathbb{R}^n$ such that $Av = \lambda v$. The vector v is then an **eigenvector** of A corresponding to λ . (The terms **characteristic vector** and **characteristic value** or **proper vector** and **proper value** are also used in place of **eigenvector** and **eigenvalue**, respectively.)

For example, the computations

$$\begin{bmatrix} 2 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

show that the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix} 2 & 2 \\ 3 & 1 \end{bmatrix}$ corresponding to the eigenvalue 4.

For many matrices A , the computation of $A^k \mathbf{x}$ for a *general vector* $\mathbf{x}$ is greatly simplified by first finding all eigenvalues and eigenvectors of A . Suppose that we can find a basis $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ for $\mathbb{R}^n$ consisting of eigenvectors of A , so that for eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ we have

$$A\mathbf{b}_i = \lambda_i \mathbf{b}_i \quad \text{for } i = 1, 2, \dots, n.$$

To compute $A^k \mathbf{x}$, we first express $\mathbf{x}$ as a linear combination of these basis eigenvectors:

$$\mathbf{x} = d_1 \mathbf{b}_1 + d_2 \mathbf{b}_2 + \dots + d_n \mathbf{b}_n.$$

Because $A^k \mathbf{b}_i = \lambda_i^k \mathbf{b}_i$, we then obtain

$$A^k \mathbf{x} = \lambda_1^k d_1 \mathbf{b}_1 + \lambda_2^k d_2 \mathbf{b}_2 + \dots + \lambda_n^k d_n \mathbf{b}_n. \quad (6)$$

It can be shown that if complex numbers are allowed as vector components and scalars, then such a basis consisting of eigenvectors usually exists, in the sense that if A is chosen at random, it exists with probability 1. When we limit ourselves to real-number components and scalars, such a basis often does not exist. However, it can be shown that if A is a real *symmetric* matrix, which is the case for some applications, then we can find a basis of eigenvectors. This idea is explored further in Section 5.2.

Equation (6) can give important information on the effect of repeatedly transforming $\mathbb{R}^n$ using multiplication by A . Suppose that at least one of the eigenvalues λ_i —we may as well assume that it is λ_1 —has magnitude greater than 1. Then vectors of arbitrarily large magnitude can be obtained under repeated multiplication of $\mathbf{b}_1$ by A —that is, $\lim_{k \rightarrow \infty} |A^k \mathbf{b}_1| = \infty$. Multiplication by A is then called an **unstable transformation** of $\mathbb{R}^n$. On the other hand, if $|\lambda_i| < 1$ for $i = 1, 2, \dots, n$, then $\lim_{k \rightarrow \infty} |A^k \mathbf{x}| = 0$ for all $\mathbf{x}$ in $\mathbb{R}^n$ and the transformation is **stable**. If the maximum of the $|\lambda_i|$ is 1, the transformation is **neutrally stable**.

Suppose now that in Eq. (6), one of the λ_i —again, we may as well assume that it is λ_1 —is of magnitude greater than all of the other λ_i . Then λ_1^k *dominates* the other λ_i^k for large values of k . If $d_1 \neq 0$, this means that the term $\lambda_1^k d_1 \mathbf{b}_1$ dominates the other terms in Eq. (6), so that for large values of k , the vector $A^k \mathbf{x}$ is nearly parallel to $\mathbf{b}_1$, and one more multiplication of $A^k \mathbf{x}$ by A amounts approximately to multiplying each component of $A^k \mathbf{x}$ by λ_1 . Indeed, one way to compute such a dominant eigenvalue λ_1 is to use a computer and find the ratio of the components of $A^{k+1} \mathbf{x}$ to those of $A^k \mathbf{x}$ for a suitable value of k . Quite small values of k will serve if λ_1 dominates strongly—for instance, if

its magnitude is several times the next largest magnitude of an eigenvalue. If the ratio of those magnitudes is closer to 1—say, 1.2—then we have to use a larger value for k . MATLAB can be used to illustrate this. (See the MATLAB exercises.) This method of computing the eigenvalue of maximum magnitude is called the *power method*, and is refined and treated in more detail in Section 8.4, where some additional methods for computing eigenvalues are described.

Computing Eigenvalues

In this section we show how a determinant can be used to find eigenvalues of an $n \times n$ matrix A ; the computational technique is practical only for relatively small matrices.

We write the equation $Av = \lambda v$ as $Av - \lambda v = 0$, or as $Av - \lambda Iv = 0$, where I is the $n \times n$ identity matrix. This last equation can be written as $(A - \lambda I)v = 0$, so v must be a solution of the homogeneous linear system

$$(A - \lambda I)x = 0. \quad (7)$$

An eigenvalue of A is thus a scalar λ for which system (7) has a nontrivial solution v . (Recall that an eigenvector is nonzero by definition.) We know that system (7) has a nontrivial solution precisely when the determinant of the coefficient matrix is zero—that is, if and only if

$$\det(A - \lambda I) = 0. \quad (8)$$

HISTORICAL NOTE THE CONCEPT OF AN EIGENVALUE, in its origins and its later development, was independent of matrix theory. In fact, its original context was that of the solution of systems of linear differential equations with constant coefficients (see Section 5.3). Jean Le Rond D'Alembert (1717–1783), in his work in the 1740s and 1750s on the motion of a string loaded with a finite number of masses, considered the system

$$\frac{d^2 y_i}{dt^2} + \sum_{k=1}^3 a_{ik} y_k = 0, \quad i = 1, 2, 3.$$

(Here the number of masses is restricted to 3 for simplicity.) To solve this system, D'Alembert multiplied the i th equation by a constant v_i (to be determined) for each i and added the equations together to obtain

$$\sum_{i=1}^3 v_i \frac{d^2 y_i}{dt^2} + \sum_{i,k=1}^3 v_i a_{ik} y_k = 0.$$

If the v_i are chosen so that $\sum_i v_i a_{ki} + \lambda v_k = 0$ for $k = 1, 2, 3$ —that is, if the vector $[v_1, v_2, v_3]$ is an eigenvector corresponding to the eigenvalue $-\lambda$ for the matrix $A = [a_{ki}]$, then the substitution $u = v_1 y_1 + v_2 y_2 + v_3 y_3$ converts the original system to the single differential equation

$$\frac{d^2 u}{dt^2} + \lambda u = 0.$$

This equation can now easily be solved and leads to solutions for the y_k . It is not difficult to show that λ is determined by a cubic equation that has three roots.

Eigenvalues also appeared in other situations involving systems of differential equations, including physical situations studied by Euler and Lagrange.

If $A = [a_{ij}]$, then the previous equation can be written

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0. \quad (9)$$

If we expand the determinant in Eq. (9), we obtain a polynomial expression $p(\lambda)$ of degree n with coefficients involving the a_{ij} . That is,

$$\det(A - \lambda I) = p(\lambda).$$

The polynomial $p(\lambda)$ is the **characteristic polynomial*** of the matrix A . The eigenvalues of A are precisely the solutions of the **characteristic equation** $p(\lambda) = 0$.

EXAMPLE 2 Find the eigenvalues of the matrix

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}.$$

SOLUTION The characteristic polynomial of A is

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 2 \\ 2 & -\lambda \end{vmatrix} = \lambda^2 - 3\lambda - 4.$$

The characteristic equation is

$$\lambda^2 - 3\lambda - 4 = 0,$$

and we obtain $(\lambda - 4)(\lambda + 1) = 0$; therefore, $\lambda_1 = -1$ and $\lambda_2 = 4$ are eigenvalues of A . ■

EXAMPLE 3 Show that $\lambda_1 = 1$ is an eigenvalue of the transition matrix for any Markov chain.

SOLUTION Let T be an $n \times n$ transition matrix for a Markov chain; that is, all entries in T are nonnegative and the sum of the entries in each column of T is 1. We easily see that the sum of the entries in any column of $T - I$ must be zero. Thus the

*Many authors define the characteristic polynomial of A to be $p(\lambda) = \det(\lambda I - A)$ rather than $p(\lambda) = \det(A - \lambda I)$. Because $\lambda I - A = (-1)(A - \lambda I)$, we see that for an $n \times n$ matrix A , we have $\det(\lambda I - A) = (-1)^n \det(A - \lambda I)$. Thus the two definitions differ only when n is an odd integer, in which case the two polynomials differ only in sign. The definition $\det(\lambda I - A)$ has the advantage that the term of highest degree is always λ^n , rather than being $-\lambda^n$ when n is odd. We use the definition $p(\lambda) = \det(A - \lambda I)$ in this first course because for our pencil-and-paper computations, it is easier to write “ $-\lambda$ ” after each diagonal entry than to change the sign of every entry and then write “ $\lambda +$ ” before the diagonal entries. However, the command `poly(A)` in MATLAB will produce the coefficients of the polynomial $\det(\lambda I - A)$.

sum of the row vectors of $T - I$ is the zero vector, so the rows of $T - I$ are linearly dependent. Consequently, $\text{rank}(T - I) < n$, and $(T - I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, so $\lambda_1 = 1$ is an eigenvalue of T . ■

The characteristic equation of an $n \times n$ matrix is a polynomial equation of degree n . This equation has n solutions if we allow both real and complex numbers and if we count the possible multiplicities greater than 1 of some solutions. Linear algebra can be done using complex scalars as well as real scalars. Introducing complex scalars makes the theory simpler and illuminates behavior in the real-scalar case. However, pencil-and-paper computations involving complex numbers can be very laborious. We can expect an eigenvector corresponding to an eigenvalue $\lambda = a + bi$ with $b \neq 0$ to have some components that are not real numbers. Exercise 44 and some of the computer exercises require computations using complex numbers.

We have now run into complex numbers while attempting to do linear algebra within $\mathbb{R}^n$. With the exception of the exercises just mentioned, we leave the computation of complex eigenvalues and of eigenvectors with complex components to Chapter 9. It is only in the context of complex n -space $\mathbb{C}^n$, which consists of all n -tuples of complex numbers, that the complete eigenstory can be explained. For example, diagonalization of a square matrix with randomly chosen real number entries, which we will discuss in Section 5.2, often cannot be achieved if we restrict ourselves only to real numbers. Using complex numbers, such a matrix can be diagonalized with probability 1. We believe it is important that you be aware of the complete eigenstory, even if you do not study Chapter 9. For this reason, theorems and definitions in this chapter are often phrased using a parenthetical *possibly complex* or stated in terms of n -space, meaning either $\mathbb{R}^n$ or $\mathbb{C}^n$. Looking back at Definition 5.1, the same definitions of eigenvalues and eigenvectors apply if A has complex entries and we allow vectors $\mathbf{v}$ in $\mathbb{C}^n$ and complex scalars λ .

The characteristic polynomial of a matrix A may have multiple roots; perhaps it has -2 as a root of multiplicity 1, and 5 as a root of multiplicity 2, corresponding to factors $(\lambda + 2)(\lambda - 5)^2$ of the characteristic polynomial. Suppose that these are the only roots. We will say that the eigenvalues are $\lambda_1 = -2$ and $\lambda_2 = \lambda_3 = 5$. That is, if there are a total of k distinct roots of the characteristic polynomial of degree n , it is convenient for us to denote the roots by $\lambda_1, \lambda_2, \dots, \lambda_k$. Our next example illustrates this.

EXAMPLE 4 Find the eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix}.$$

SOLUTION The characteristic polynomial is

$$p(\lambda) = \begin{vmatrix} 2 - \lambda & 1 & 0 \\ -1 & -\lambda & 1 \\ 1 & 3 & 1 - \lambda \end{vmatrix} = (2 - \lambda) \begin{vmatrix} -\lambda & 1 \\ 3 & 1 - \lambda \end{vmatrix} - 1 \begin{vmatrix} -1 & 1 \\ 1 & 1 - \lambda \end{vmatrix}$$

$$\begin{aligned}
 &= (2 - \lambda)(\lambda^2 - \lambda - 3) - (\lambda - 2) = -(\lambda - 2)(\lambda^2 - \lambda - 2) \\
 &= -(\lambda - 2)(\lambda + 2)(\lambda + 1).
 \end{aligned}$$

Hence, the eigenvalues of A are $\lambda_1 = -1$ and $\lambda_2 = \lambda_3 = 2$. ■

Computation of Eigenvectors

We turn to the computation of the eigenvectors corresponding to an eigenvalue λ of a matrix A . Having found the eigenvalue, we substitute it in homogeneous system (7) and solve to find the nontrivial solutions of the system. We will obtain an infinite number of nontrivial solutions, each of which is an eigenvector corresponding to the eigenvalue λ .

EXAMPLE 5 Find the eigenvectors corresponding to each eigenvalue found in Example 4 for the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix}.$$

SOLUTION The eigenvalues of A were found to be $\lambda_1 = -1$ and $\lambda_2 = \lambda_3 = 2$. We substitute each of these values in the homogeneous system (7). The eigenvectors are obtained by reducing the coefficient matrix $A - \lambda I$ in the augmented matrix for the system. For $\lambda_1 = -1$, we obtain

$$\begin{aligned}
 [A - \lambda_1 I \mid 0] &= [A + I \mid 0] \\
 &= \left[\begin{array}{ccc|c} 3 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & 4 & 3 & 0 \\ 0 & -8 & -6 & 0 \end{array} \right] \\
 &\sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].
 \end{aligned}$$

(In the future, we will drop the column of zeros to the right of the partition when solving for eigenvectors.) The solution of the homogeneous system is given by

$$\begin{bmatrix} r/4 \\ -3r/4 \\ r \end{bmatrix} \text{ for any scalar } r.$$

Therefore,

$$v_1 = \begin{bmatrix} r/4 \\ -3r/4 \\ r \end{bmatrix} = r \begin{bmatrix} 1/4 \\ -3/4 \\ 1 \end{bmatrix} \text{ for any nonzero scalar } r$$

is an eigenvector corresponding to the eigenvalue $\lambda_1 = -1$. Replacing r by $4r$, we can express this result without fractions as

$$\mathbf{v}_1 = \begin{bmatrix} r \\ -3r \\ 4r \end{bmatrix} = r \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \quad \text{for any nonzero scalar } r.$$

For $\lambda_2 = 2$, we obtain

$$\begin{aligned} [A - \lambda_2 I] &= [A - 2I] \\ &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & -2 & 1 \\ 1 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

This time we find that

$$\mathbf{v}_2 = \begin{bmatrix} s \\ 0 \\ s \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{for any nonzero scalar } s$$

is an eigenvector. As a check, we could compute $A\mathbf{v}_1$ and $A\mathbf{v}_2$. For example, we have

$$A\mathbf{v}_2 = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} s \\ 0 \\ s \end{bmatrix} = \begin{bmatrix} 2s \\ 0 \\ 2s \end{bmatrix} = 2 \begin{bmatrix} s \\ 0 \\ s \end{bmatrix} = 2\mathbf{v}_2 = \lambda_2 \mathbf{v}_2.$$

EXAMPLE 6 Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -6 \\ 8 & 1 & 9 \end{bmatrix}.$$

SOLUTION The characteristic polynomial of A is

$$\begin{aligned} p(\lambda) &= |A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ -8 & 4-\lambda & -6 \\ 8 & 1 & 9-\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} 4-\lambda & -6 \\ 1 & 9-\lambda \end{vmatrix} \\ &= (1-\lambda)(\lambda^2 - 13\lambda + 42) = (1-\lambda)(\lambda-6)(\lambda-7). \end{aligned}$$

The eigenvalues of A are $\lambda_1 = 1$, $\lambda_2 = 6$, and $\lambda_3 = 7$. For $\lambda_1 = 1$, we have

$$A - \lambda_1 I = A - I = \begin{bmatrix} 0 & 0 & 0 \\ -8 & 3 & -6 \\ 8 & 1 & 8 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 2 \\ 8 & 1 & 8 \end{bmatrix}$$

$$\sim \begin{bmatrix} 8 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{8} & 1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{15}{16} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix},$$

so

$$\mathbf{v}_1 = \begin{bmatrix} -15r/16 \\ -r/2 \\ r \end{bmatrix} = r \begin{bmatrix} -15/16 \\ -1/2 \\ 1 \end{bmatrix} \text{ for any nonzero scalar } r$$

is an eigenvector. Replacing r by $-16r$, we can express this result without the fractions as

$$\mathbf{v}_1 = \begin{bmatrix} 15r \\ 8r \\ -16r \end{bmatrix} = r \begin{bmatrix} 15 \\ 8 \\ -16 \end{bmatrix} \text{ for any nonzero scalar } r.$$

For $\lambda_2 = 6$, we have

$$A - \lambda_2 I = A - 6I = \begin{bmatrix} -5 & 0 & 0 \\ -8 & -2 & -6 \\ 8 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -6 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix},$$

so

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ -3s \\ s \end{bmatrix} = s \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix} \text{ for any nonzero scalar } s$$

is an eigenvector. Finally, for $\lambda_3 = 7$, we have

$$A - \lambda_3 I = A - 7I = \begin{bmatrix} -6 & 0 & 0 \\ -8 & -3 & -6 \\ 8 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & -6 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix},$$

so

$$\mathbf{v}_3 = \begin{bmatrix} 0 \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \text{ for any nonzero scalar } t$$

is an eigenvector. ■

Properties of Eigenvalues and Eigenvectors

We turn now to algebraic properties of eigenvalues and eigenvectors. The properties given in Theorem 5.1 are so straightforward and instructive to prove that we leave the proofs as Exercises 27–29.

THEOREM 5.1 Properties of Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

1. If λ is an eigenvalue of A with $\mathbf{v}$ as a corresponding eigenvector, then λ^k is an eigenvalue of A^k , again with $\mathbf{v}$ as a corresponding eigenvector, for any positive integer k .
2. If λ is an eigenvalue of an invertible matrix A with $\mathbf{v}$ as a corresponding eigenvector, then $\lambda \neq 0$ and $1/\lambda$ is an eigenvalue of A^{-1} , again with $\mathbf{v}$ as a corresponding eigenvector.
3. If λ is an eigenvalue of A , then the set E_λ consisting of the zero vector together with all eigenvectors of A for this eigenvalue λ is a subspace of n -space, the **eigenspace** of λ .

Eigenvalues and Transformations

Let us consider the significance of an eigenvalue λ and corresponding eigenvector $\mathbf{v}$ of an $n \times n$ matrix A for the associated linear transformation $T(\mathbf{x}) = A\mathbf{x}$. The equation

$$A\mathbf{v} = \lambda\mathbf{v}$$

takes the form

$$T(\mathbf{v}) = \lambda\mathbf{v}.$$

Thus, the linear transformation T maps the vector $\mathbf{v}$ onto a vector that is parallel to $\mathbf{v}$. (See Fig. 5.1.) We present a definition of eigenvalues and eigenvectors for linear transformations that is more general than for matrices, in that we need not restrict ourselves to a finite-dimensional vector space.

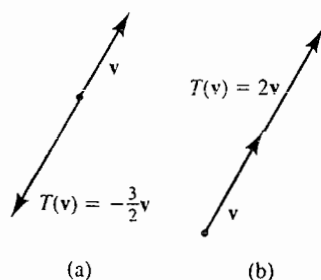


FIGURE 5.1

(a) T has eigenvalue $\lambda = -\frac{3}{2}$; (b) T has eigenvalue $\lambda = 2$.

DEFINITION 5.2 Eigenvalues and Eigenvectors

Let T be a linear transformation of a vector space V into itself. A scalar λ is an **eigenvalue** of T if there is a *nonzero* vector $\mathbf{v}$ in V such that $T(\mathbf{v}) = \lambda\mathbf{v}$. The vector $\mathbf{v}$ is then an **eigenvector** of T corresponding to λ .

It is significant that we can define eigenvalues and eigenvectors for a linear transformation $T: V \rightarrow V$ without any reference to a matrix representation and without even assuming that V is finite-dimensional. Example 8 will discuss the eigenvalues and eigenvectors of a linear transformation that lies at the heart of calculus and deals with an infinite-dimensional vector space. Students who have studied exponential growth problems in calculus will recognize the importance of this example.

ILLUSTRATION 1

Not every linear transformation has eigenvectors. Rotation of the plane counterclockwise through a positive angle θ is a linear transformation (see page 156). If $0 < \theta < 180^\circ$, then no vector is mapped onto one parallel to it—that is, no vector is an eigenvector. If $\theta = 180^\circ$, then every nonzero vector is an eigenvector and they all have the same associated eigenvalue $\lambda_1 = -1$. ■

ILLUSTRATION 2

The linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that reflects vectors in the line $x + 2y = 0$ maps the vector $[2, -1]$ onto itself and maps the vector $[1, 2]$ onto $[-1, -2]$, as indicated in Figure 5.2. The equations $T([2, -1]) = [2, -1]$ and $T([1, 2]) = [-1, -2]$ show that $[2, -1]$ and $[1, 2]$ are eigenvectors of T with corresponding eigenvalues 1 and -1 , respectively. ■

For a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$, we can find the transformation's eigenvalues and eigenvectors by finding those of its standard matrix representation. The following example illustrates how this works.

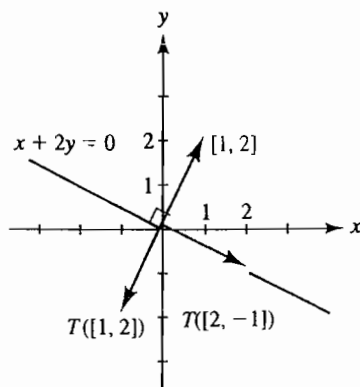


FIGURE 5.2
Reflection in the line $x + 2y = 0$.

EXAMPLE 7 Find the eigenvalues λ and eigenvectors $\mathbf{v}$ of the linear transformation: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x_1, x_2, x_3]) = [x_1, -8x_1 + 4x_2 - 6x_3, 8x_1 + x_2 + 9x_3]$. Illustrate the equation $T(\mathbf{v}) = \lambda\mathbf{v}$ for each eigenvalue.

SOLUTION Writing vectors as column vectors, we may express the linear transformation T as $T(\mathbf{x}) = A\mathbf{x}$, where A is the matrix given in Example 6. The rest of the solution proceeds precisely as in that example. Let us return to row notation and illustrate the action of T on the basic eigenvectors obtained by taking $r = s = t = 1$ in the expressions for $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ in Example 6. We have

$$\begin{aligned} T([15, 8, -16]) &= [15, -8(15) + 4(8) - 6(-16), 8(15) + 8 + 9(-16)] \\ &= [15, 8, -16], \end{aligned}$$

$$\begin{aligned} T([0, -3, 1]) &= [0, -8(0) + 4(-3) - 6(1), 8(0) - 3 + 9(1)] \\ &= [0, -18, 6] = 6[0, -3, 1], \end{aligned}$$

$$\begin{aligned} T([0, -2, 1]) &= [0, -8(0) + 4(-2) - 6(1), 8(0) - 2 + 9(1)] \\ &= [0, -14, 7] = 7[0, -2, 1]. \end{aligned}$$

We now give an example from calculus, involving a vector space that is not finite-dimensional.

EXAMPLE 8 Let D_x be the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ and having derivatives of all orders. Let $T: D_x \rightarrow D_x$ be the differentiation map, so that $T(f) = f'$, the derivative of f . Describe all eigenvalues and eigenvectors of T . (We have seen in Section 3.4 that differentiation does give a linear transformation.)

SOLUTION We must find scalars λ and nonzero functions f such that $T(f) = \lambda f$ —that is, such that $f' = \lambda f$. We consider two cases: if $\lambda = 0$, and if $\lambda \neq 0$.

If $\lambda = 0$, we are trying to solve the differential equation $f' = 0$, or to use Leibniz notation, $dy/dx = 0$. We know from calculus that the only solutions of this equation are the constant functions. Thus the nonzero constant functions are eigenvectors corresponding to the eigenvalue 0.

If $\lambda \neq 0$, the differential equation becomes $f' = \lambda f$ or $dy/dx = \lambda y$. It is readily checked that $y = e^{\lambda x}$ is a solution of this equation, so $f(x) = ke^{\lambda x}$ is an eigenvector for every nonzero scalar k . To see that these are the only solutions, we can solve the differential equation by separating variables, which yields the equation

$$\frac{dy}{y} = \lambda dx.$$

Integrating both sides of the equation yields

$$\ln|y| = \lambda x + c.$$

Solving for y , we obtain $y = \pm e^c e^{\lambda x} = ke^{\lambda x}$, so the only solutions of the differential equation are indeed of the form $y = ke^{\lambda x}$. ■

ILLUSTRATION 3 There are several physical situations in which a point of a vibrating body is subject to a restoring force proportional to its displacement from its position at rest. One such situation is the vibration of a body suspended by a spring, as

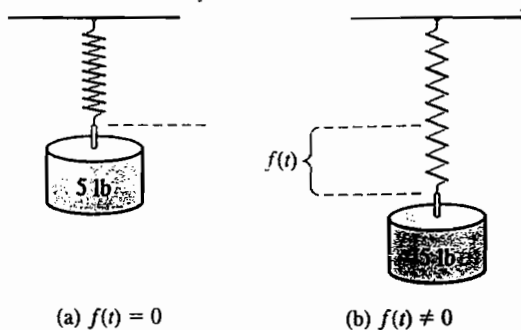


FIGURE 5.3
A vibrating body suspended by a spring.

illustrated in Figure 5.3. We let $f(t)$ be the displacement of the body at time t . The acceleration of the body at time t is then given by $f''(t)$. Using Newton's law $F = ma$, we see that we can express the relation between the restoring force and the displacement as $mf''(t) = -cf(t)$, where c is a positive constant. Dividing by m , we can rewrite this differential equation as

$$f''(t) = -k^2 f(t) \quad (10)$$

for some constant k . Now differentiating twice is a linear transformation of the vector space of all infinitely differentiable function into itself, and we see that the equation $f''(t) = -k^2 f(t)$ asserts that the function f is an eigenvector of this transformation with eigenvalue $-k^2$. We can easily check that the functions $\sin kt$ and $\cos kt$ are solutions of Eq. (10). By property 3 of Theorem 5.1, every linear combination

$$f(t) = a \sin kt + b \cos kt$$

is also a solution. Note that for this vibration situation, the eigenvalue $-k^2$ determines the *frequency* of the vibration. To illustrate, suppose in the case of the spring in Figure 5.3 we have $f(t) = 2$ and $f'(t) = 0$ when $t = 0$. It is readily determined that we then must have $b = 2$ and $a = 0$, so $f(t) = 2 \cos kt$. Thus the frequency of vibration, which is the reciprocal of the period, is $k/(2\pi)$. ■

SUMMARY

Let A be an $n \times n$ matrix.

1. If $A\mathbf{v} = \lambda\mathbf{v}$, where $\mathbf{v}$ is a nonzero column vector and λ is a scalar, then λ is an eigenvalue of A and $\mathbf{v}$ is an eigenvector of A corresponding to λ .
2. The characteristic polynomial $p(\lambda)$ of A is obtained by expanding the determinant $|A - \lambda I|$, where I is the $n \times n$ identity matrix.
3. The eigenvalues λ of A can be found by solving the characteristic equation $p(\lambda) = |A - \lambda I| = 0$. There are at most n real solutions λ of this equation.

- The eigenvectors of A corresponding to λ are the nontrivial solutions of the homogeneous system $(A - \lambda I)\mathbf{x} = \mathbf{0}$, as illustrated in Examples 5 and 6.
- Let k be a positive integer. If λ is an eigenvalue of A having $\mathbf{v}$ as eigenvector, then λ^k is an eigenvalue of A^k with $\mathbf{v}$ as eigenvector. If A is invertible, this statement is also true for $k = -1$.
- Let λ be an eigenvalue of A . The set E_λ in n -space consisting of the zero vector and all eigenvectors corresponding to λ is a subspace of n -space.
- Let T be a linear transformation of a vector space V into itself. A nonzero $\mathbf{v}$ in V is an eigenvector of T if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some scalar λ , which is called the eigenvalue of T corresponding to $\mathbf{v}$.
- If T is a linear transformation of the vector space $\mathbb{R}^n$ into itself, the eigenvalues of T are the eigenvalues of the standard matrix representation of T .

EXERCISES

1. Consider the matrices

$$A_1 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

and the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix},$$

$$\mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

List the vectors that are eigenvectors of A_1 and the ones that are eigenvectors of A_2 . Give the eigenvalue in each case.

8. $\begin{bmatrix} -1 & 0 & 0 \\ -4 & 2 & -1 \\ 4 & 0 & 3 \end{bmatrix}$

9. $\begin{bmatrix} 8 & 0 & 0 \\ 7 & -1 & -2 \\ -7 & 0 & 1 \end{bmatrix}$

10. $\begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$

11. $\begin{bmatrix} -2 & 0 & 0 \\ -5 & -2 & -5 \\ 5 & 0 & 3 \end{bmatrix}$

12. $\begin{bmatrix} -4 & 0 & 0 \\ -7 & 2 & -1 \\ 7 & 0 & 3 \end{bmatrix}$

13. $\begin{bmatrix} -1 & 0 & 1 \\ -7 & 2 & 5 \\ 3 & 0 & 1 \end{bmatrix}$

14. $\begin{bmatrix} 4 & 0 & 0 \\ 8 & 4 & 8 \\ 0 & 0 & 4 \end{bmatrix}$

15. $\begin{bmatrix} 0 & 0 & 1 \\ -2 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$

16. $\begin{bmatrix} 2 & 0 & 1 \\ 6 & 4 & -3 \\ 2 & 0 & 3 \end{bmatrix}$

In Exercises 2–16, find the characteristic polynomial, the real eigenvalues, and the corresponding eigenvectors of the given matrix.

2. $\begin{bmatrix} 7 & 5 \\ -10 & -8 \end{bmatrix}$

3. $\begin{bmatrix} -1 & -2 \\ 4 & 5 \end{bmatrix}$

4. $\begin{bmatrix} -7 & -5 \\ 16 & 17 \end{bmatrix}$

5. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix}$

7. $\begin{bmatrix} 2 & 0 & 0 \\ 1 & -1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$

In Exercises 17–22, find the eigenvalues λ_i and the corresponding eigenvectors $\mathbf{v}_i$ of the linear transformation T .

17. T defined on $\mathbb{R}^2$ by $T([x, y]) = [2x - 3y, -3x + 2y]$

18. T defined on $\mathbb{R}^2$ by $T([x, y]) = [x - y, -x + y]$

19. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [x_1 + x_3, x_2, x_1 + x_3]$

10. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [x_1, 4x_2 + 7x_3, 2x_2 - x_3]$

11. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [x_1, -5x_1 + 3x_2 - 5x_3, -3x_1 - 2x_2]$

12. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [3x_1 - x_2 + x_3, -2x_1 + 2x_2 - x_3, 2x_1 + x_2 + 4x_3]$

13. Mark each of the following True or False.

a. Every square matrix has real eigenvalues.

b. Every $n \times n$ matrix has n distinct (possibly complex) eigenvalues.

c. Every $n \times n$ matrix has n not necessarily distinct and possibly complex eigenvalues.

d. There can be only one eigenvalue associated with an eigenvector of a linear transformation.

e. There can be only one eigenvector associated with an eigenvalue of a linear transformation.

f. If $\mathbf{v}$ is an eigenvector of a matrix A , then $\mathbf{v}$ is an eigenvector of $A + cI$ for all scalars c .

g. If λ is an eigenvalue of a matrix A , then λ is an eigenvalue of $A + cI$ for all scalars c .

h. If $\mathbf{v}$ is an eigenvector of an invertible matrix A , then $c\mathbf{v}$ is an eigenvector of A^{-1} for all nonzero scalars c .

i. Every vector in a vector space V is an eigenvector of the identity transformation of V into V .

j. Every nonzero vector in a vector space V is an eigenvector of the identity transformation of V into V .

24. Let $T: V \rightarrow V$ be a linear transformation of a vector space V into itself, and let λ be a scalar. Prove that $\{\mathbf{v} \in V \mid T(\mathbf{v}) = \lambda\mathbf{v}\}$ is a subspace of V .

25. Prove that if A is a square matrix, then AA^T and A^TA have the same eigenvalues.

26. Following the idea in Example 8, show that the functions e^{ax} , e^{-ax} , $\sin ax$, and $\cos ax$ are eigenvectors for the linear transformation $T: D_\infty \rightarrow D_\infty$ defined by $T(f) = f^{(4)}$, the fourth derivative of f . Indicate the eigenvalue in each case.

27. Prove property 1 of Theorem 5.1.

28. Prove property 2 of Theorem 5.1.

29. Prove property 3 of Theorem 5.1.

30. Prove that a square matrix is invertible if and only if no eigenvalue is zero.

31. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

used to generate the Fibonacci sequence (2).

32. Let A be an $n \times n$ matrix and let I be the $n \times n$ identity matrix. Compare the eigenvectors and eigenvalues of A with those of $A + rI$ for a scalar r .

33. Let a square matrix A with real eigenvalues have a unique eigenvalue of greatest magnitude. Numerical computation of this eigenvalue by the *power method* (discussed in Section 8.4) can be difficult if there is another eigenvalue of almost equal magnitude, so the ratio of these magnitudes is close to 1.

- Suppose we know that a 4×4 matrix A has eigenvalues of approximately 20, 2, -3, and -19.5. Using Exercise 32, how might we modify A so that the aforementioned ratio is not so close to 1?
- Repeat part (a), given that the eigenvalues are known to be approximately 19.5, 2, -3, and -20.

34. Let A be an $n \times n$ real matrix. An eigenvector $\mathbf{w}$ in $\mathbb{R}^n$ and a corresponding eigenvalue α of A^T are also called a *left eigenvector and eigenvalue* of A . Explain the reason for this name.

35. (*Principle of biorthogonality*) Let A be an $n \times n$ real matrix. Let $\mathbf{v}$ in $\mathbb{R}^n$ be an eigenvector of A with corresponding eigenvalue λ , and let $\mathbf{w}$ in $\mathbb{R}^n$ be an eigenvector of A^T with corresponding eigenvalue α . Prove that if $\lambda \neq \alpha$, then $\mathbf{v}$ and $\mathbf{w}$ are perpendicular vectors. [HINT: Refer to Exercise 34, and compute $\mathbf{w}^T A \mathbf{v}$ in two ways, using associativity of matrix multiplication.]

- Prove that the eigenvalues of an $n \times n$ real matrix A are the same as the eigenvalues of A^T .
- With reference to part (a), show by a counterexample that an eigenvector of A need not be an eigenvector of A^T .

37. The trace of an
- $n \times n$
- matrix
- A
- is defined by

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Let the characteristic polynomial $p(\lambda)$ factor into linear factors, so that A has n (not necessarily distinct) eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Prove that

$$\begin{aligned}\operatorname{tr}(A) &= (-1)^{n-1}(\text{Coefficient of } \lambda^{n-1} \text{ in } p(\lambda)) \\ &= \lambda_1 + \lambda_2 + \cdots + \lambda_n.\end{aligned}$$

38. Let A be an $n \times n$ matrix, and let C be an invertible $n \times n$ matrix. Show that the eigenvalues of A and of $C^{-1}AC$ are the same. [HINT: Show that the characteristic polynomials of the two matrices are the same.]

39. *Cayley–Hamilton theorem: Every square matrix A satisfies its characteristic equation. That is, if the characteristic equation is $p_n\lambda^n + p_{n-1}\lambda^{n-1} + \cdots + p_1\lambda + p_0 = 0$, then $p_nA^n + p_{n-1}A^{n-1} + \cdots + p_1A + p_0I = O$, the zero matrix. Illustrate the Cayley–Hamilton theorem for the matrix $\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$.*

40. Let A be an invertible $n \times n$ matrix. Using the Cayley–Hamilton theorem stated in the preceding exercise, show that A^{-1} can be computed as a linear combination of the powers $A, A^2, \dots, A^n$ of A . Compute the inverse of the matrix A in the preceding exercise in this fashion.
41. Let v_1 and v_2 be eigenvectors of a linear transformation $T: V \rightarrow V$ with corresponding eigenvalues λ_1 and λ_2 , respectively. Prove that, if $\lambda_1 \neq \lambda_2$, then v_1 and v_2 are independent vectors.
42. The analogue of Exercise 41 for a list of r eigenvectors in V having distinct eigenvalues is also true; that is, the vectors are independent. See if you can prove it. [HINT: Suppose that the vectors are dependent; consider the first vector in the list that is a linear combination of its predecessors, and apply T .]
43. State the result for matrices corresponding to Exercise 42. Explain why successful completion of Exercise 42 gives a proof of this statement for matrices.

44. Let

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

It can be shown that, if x is any column vector in $\mathbb{R}^2$, then Ax can be obtained geometrically from x by rotating x counterclockwise through an angle of 90° . For example, we find that $Ae_1 = e_2$ and $Ae_2 = -e_1$.

- Argue geometrically that A has no real eigenvalues.
 - Find the complex eigenvalues and eigenvectors of A .
 - Argue geometrically that A^2 should have real eigenvalues _____. (Fill in the blank.)
 - Use part (b) and Theorem 5.1 to find the eigenvalues of A^2 , and compare with the answer obtained for part (c). What are the real eigenvectors of A^2 ? the complex eigenvectors of A^2 ?
 - Find eigenvalues and eigenvectors for A^3 .
 - Find eigenvalues and eigenvectors for A^4 .
45. Use MATCOMP in LINTEK, or MATLAB, and relation (4) to find the following terms of the Fibonacci sequence (2) as accurately as possible. (Use double-precision printing if possible.)
- F_8 (Note that $F_8 = 21$; this part is to check procedure.)
 - F_{30}
 - F_{50}
 - F_{77}
 - F_{150}
46. The first two terms of a sequence are $a_0 = 0$ and $a_1 = 1$. Subsequent terms are generated using the relation
- $$a_k = 2a_{k-1} + a_{k-2} \quad \text{for } k \geq 2.$$
- Write the terms of the sequence through a_8 .
 - Find a matrix that can be used to generate the sequence, as the matrix A in Exercise 31 can be used to generate the Fibonacci sequence.
 - Use MATCOMP in LINTEK, or MATLAB, to find a_{30} .
47. Repeat Exercise 46 for a sequence where $a_0 = 0, a_1 = 1, a_2 = 2$, and $a_k = 2a_{k-1} - 3a_{k-2} + a_{k-3}$ for $k \geq 3$.

For a square matrix A , the MATLAB command `eig(A)` produces the eigenvalues (both real and complex) of A . The command `[V, D] = eig(A)` produces a matrix V whose column vectors are (perhaps complex) eigenvectors of A and a diagonal matrix D whose entry d_{ii} is the eigenvalue for the eigenvector in the i th column of V . In Exercises 48–51, use either MATLAB or the routine `MATCOMP` in `LINTEK` to find the real eigenvalues and corresponding eigenvectors of the given matrix.

$$48. \begin{bmatrix} 7 & 10 & 6 \\ 2 & -1 & -6 \\ -2 & -5 & 0 \end{bmatrix} \quad 49. \begin{bmatrix} -1 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ -3 & 0 & 2 & 0 \\ -3 & 1 & 0 & 3 \end{bmatrix}$$

$$50. \begin{bmatrix} 0 & 0 & -2 & 2 \\ -3 & 4 & -3 & 3 \\ -4 & 0 & 2 & 4 \\ -2 & 0 & 2 & 4 \end{bmatrix}$$

$$51. \begin{bmatrix} 4 & 0 & 0 & 0 \\ -6 & 16 & -6 & 6 \\ -16 & 0 & 20 & 16 \\ -16 & 0 & 16 & 20 \end{bmatrix}$$

The routine `ALLROOTS` in `LINTEK` can be used to find both real and complex roots of a polynomial. The program uses Newton's method, which finds a solution by successive approximations of the polynomial function by a linear one. `ALLROOTS` is designed so that the user can watch the approximations approach a solution. Of course, a program such as MATLAB, which is designed for research, simply spits out the answers. In Exercises 52–55, either

- use the command `eig(A)` in MATLAB to find all eigenvalues of the matrix or
- first use `MATCOMP` in `LINTEK` to find the characteristic equation of the given matrix.

Copy down the equation, and then use `ALLROOTS` to find all eigenvalues of the matrix.

$$52. \begin{bmatrix} -1 & 4 & 6 \\ 2 & 7 & 9 \\ -3 & 11 & 13 \end{bmatrix} \quad 53. \begin{bmatrix} 10 & -13 & 8 \\ 3 & -20 & 5 \\ -11 & 7 & -6 \end{bmatrix}$$

$$54. \begin{bmatrix} -7 & 11 & -7 & 10 \\ 5 & 8 & -13 & 3 \\ -15 & 8 & -9 & 2 \\ 3 & -4 & 20 & -6 \end{bmatrix}$$

$$55. \begin{bmatrix} 21 & -8 & 0 & 32 \\ -14 & 17 & -6 & 9 \\ 15 & 11 & -13 & 16 \\ -18 & 30 & 43 & 31 \end{bmatrix}$$

56. Use the routine `MATCOMP` in `LINTEK`, or MATLAB, to illustrate the Cayley–Hamilton theorem for the matrix

$$\begin{bmatrix} -2 & 4 & 6 & -1 \\ 5 & -8 & 3 & 2 \\ 11 & -3 & 7 & 1 \\ 0 & -5 & 9 & 10 \end{bmatrix}$$

(See Exercise 39.)

57. The eigenvalue option of the routine `VECTGRPH` in `LINTEK` is designed to illustrate graphically, for a linear transformation of $\mathbb{R}^2$ into itself having real eigenvalues, how repeatedly transforming a vector makes it swing in the direction of an eigenvector having eigenvalue of maximum magnitude. By finding graphically a vector whose transform is parallel to it, one can estimate from the graph an eigenvector and the corresponding eigenvalue. Read the directions for this option, and then work with it until you can reliably achieve a score of 85% or better.

MATLAB

The command `v = poly(A)` in MATLAB produces a vector v whose components are the coefficients of the characteristic polynomial $p(\lambda)$ of A , appearing in order of decreasing powers of λ . The command `roots(v)` then produces the solutions of $p(\lambda) = 0$.

- M1. Enter the command `format long` and then use the commands just explained to find the characteristic polynomial and both the real and the complex eigenvalues of
- the matrix in Exercise 53
 - the matrix in Exercise 54.

[Recall that in MATLAB, the characteristic polynomial of an $n \times n$ matrix A is $\det(\lambda I - A)$, rather than $\det(A - \lambda I)$. Thus the coefficient of λ^n should be 1 rather than $(-1)^n$. We enter the long format because otherwise, with the displays in scientific notation, it might seem from MATLAB that the coefficient of λ^n is 0.]

Let A be an $n \times n$ matrix. In the introduction to this section, we considered the case where there is a basis $\{b_1, b_2, \dots, b_n\}$ for $\mathbb{R}^n$ consisting of eigenvectors of A , and where, for one of the eigenvectors, say b_1 , the corresponding eigenvalue λ_1 is of greater magnitude than each of $\lambda_2, \dots, \lambda_n$. Also if $x = d_1 b_1 + d_2 b_2 + \dots + d_n b_n$ with $d_1 \neq 0$, then for sufficiently large values of k , we saw that

$$A^{k+1}x \approx \lambda_1(A^k x),$$

so that this eigenvalue λ_1 of maximum magnitude might be estimated by taking the ratio of components of $A^{k+1}x$ to components of $A^k x$. Recall that a period before an arithmetic operation symbol, such as `.*` or `./`, will cause that operation to be performed component by component for vectors or matrices. Thus the MATLAB command

$$r = (A^{k+1}x) ./ (A^k x)$$

will cause the desired ratios of components to be printed. If these ratios agree for all components, we expect we have computed the eigenvalue of maximum magnitude accurate to all figures shown.

- M2. Enter `k = 1; x = [1 2 3]'; A = [3 5 -11; 5 -8 -3; -11 -3 14];` in MATLAB. (Recall that the final semicolon keeps the data from being displayed.) Then enter `r = (A^(k+1)*x) ./ (A^k*x)` to see the ratios displayed. Now enter `k = 5;` and use the up-arrow key to have the ratios computed for that k . Continue using the up-arrow key, setting k successively equal to 10, 15, 20, 25, ... until the ratios agree to all places shown. Then change to `format long`, and continue until you have found the eigenvalue of maximum magnitude accurate to all figures shown. Copy it down as your answer to this problem. Check it using the command `eig(A)`.
- M3. Property 2 of Theorem 5.1 indicates that the eigenvalue of minimum magnitude of A is the reciprocal of the eigenvalue of maximum magnitude of A^{-1} . Continuing the preceding exercise, enter `B = A;` to save A , and then enter `A = inv(A);`. Now enter `s = ones(3,1) ./ r.` (Enter `help ones` to learn the effect of the `ones(3,1)` statement.) Using the up-arrow key to access statements defining k and computing r and s , compute the eigenvalue of A of minimum magnitude accurate to all figures shown in the long format. Copy down your answer, and check it using the `eig(B)` command.
- M4. Raising a matrix A to a high power can generate error, and can cause overflow (numbers too large to handle) in a computer. Continuing the preceding two exercises, let us avoid this by only raising A to a low power, say the 5th power, and replace x by $A^5 x$ after each iteration. Repeating this m times should have the same effect as raising A to the power $5m$. Of course, now the entries in the vector x may get large. We compensate for this by *normalizing* x to

be of magnitude 1 before the next iteration. This does not change the direction of x , which should swing to parallel the eigenvector b_1 having eigenvalue λ_1 of maximum magnitude as the iterations are performed. To execute this procedure, enter $A = B$; to recover A as in Exercise M2, and then enter

$$x = A^5 x; x = (1/\text{norm}(x)) * x; r = (A * x) ./ x$$

to compute $A^5 x$, normalize x , and compute the ratios of components of Ax to those of x . One repetition of the up-arrow key followed by the Enter key executes these iterations rapidly. Establish your result in Exercise M2 again, and then, replacing A by A^{-1} , establish your result in Exercise M3 again.

- M5. Explain why the final vector x obtained when finding the eigenvalue of either maximum or of minimum magnitude in Exercise M4 should be an eigenvector corresponding to that eigenvalue. Check that this is so using the $[V, D] = \text{eig}(A)$ command explained before Exercise 48.

In Exercises M6–M8, use the displayed command in Exercise M3 preceded by any necessary modifications in the initial vector x to find the eigenvalues of maximum and minimum magnitude. Always reinitialize x to something like $x = [1 \ 2 \ 3]$ before finding the eigenvalue of minimum magnitude. Give answers in short format. The matrices are in the file of matrices for this section, if it is available at your installation.

$$\begin{array}{lll} \text{M6. } \begin{bmatrix} -2 & 6 & 8 \\ 6 & -3 & 2 \\ 8 & 2 & -4 \end{bmatrix} & \text{M7. } \begin{bmatrix} 4 & 11 & 5 & -7 \\ 11 & -2 & -12 & 0 \\ 5 & 12 & -3 & 8 \\ -7 & -3 & 8 & 13 \end{bmatrix} & \text{M8. } \begin{bmatrix} -4 & 7 & 8 & -11 & 5 \\ 7 & -3 & 0 & 2 & 10 \\ 8 & 0 & 2 & 1 & 4 \\ -11 & 2 & 1 & 5 & -9 \\ 5 & 10 & 4 & -9 & 3 \end{bmatrix} \end{array}$$

5.2

DIAGONALIZATION

Recall that a square matrix is called **diagonal** if all entries not on the main diagonal are zero. In the preceding section we indicated the importance of being able to compute $A^k x$ for an $n \times n$ matrix A and a column vector x in $\mathbb{R}^n$. In this section, we show that, if A has n distinct eigenvalues, then computation of A^k can be essentially replaced by computation of D^k , where D is a diagonal matrix with the eigenvalues of A as diagonal entries. Notice that D^k is the diagonal matrix obtained from D by raising each diagonal entry to the power k . For example,

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}^3 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -8 \end{bmatrix}.$$

The theorem that follows shows how to summarize the action of a square matrix on its eigenvectors in a single matrix equation. This theorem is the first step toward our goal of *diagonalizing* a matrix. Theorems stated are valid for matrices and vectors with complex entries and complex scalars unless we use the adjective *real*. We sometimes refer to *n-space* in this section, meaning either $\mathbb{R}^n$ or $\mathbb{C}^n$.

THEOREM 5.2 Matrix Summary of Eigenvalues of A

Let A be an $n \times n$ matrix and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be (possibly complex) scalars and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be nonzero vectors in n -space. Let C be the $n \times n$ matrix having $\mathbf{v}_j$ as j th column vector, and let

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}.$$

Then $AC = CD$ if and only if $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of A and $\mathbf{v}_j$ is an eigenvector of A corresponding to λ_j for $j = 1, 2, \dots, n$.

PROOF We have

$$\begin{aligned} CD &= \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix} \\ &= \begin{bmatrix} | & | & & | \\ \lambda_1 \mathbf{v}_1 & \lambda_2 \mathbf{v}_2 & \cdots & \lambda_n \mathbf{v}_n \\ | & | & & | \end{bmatrix}. \end{aligned}$$

On the other hand,

$$AC = A \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \\ | & | & & | \end{bmatrix}.$$

Thus, $AC = CD$ if and only if $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$. $\blacktriangle$